24/02/26.

Top - 5 (SVM).

Dataset :- Class (Spam = +1 ; Not Spam = -1)

Suspicious Words.	Links	Attachement	class	y
6	3	1	+1	≥ 1
1	0	0	-1	≥ 1
4	2	1	+1	≥ 1
0	0	0	-1	≥ 1
5	2	1	+1	≥ 1

Solution:-

SW (x_1) → w_1 Links (x_2) → w_2 Attach (x_3) → w_3 class (Bias : w_0)

Weighted Sum:

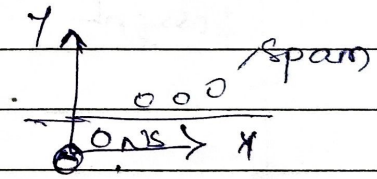
$f(x) = z = w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3$

Initialize:

w_0 & $w_2 = 0$.

Spam → (4, 5, 6) (2, 3) (1)

Non-Spam → (0, 1) (0) (0)



Spam

$= f(x) = 0$.

Non-Spam

Decision boundary

If $f(x) = 0 \Rightarrow w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3 = 0$.

$\Rightarrow w_0 + w_1 x_1 = 0$.

Let $x_1 = 2$.; Now, $w_1 = 1$.

$w_0 + w_1 x_1 = 0$.

$w_0 + (1 \cdot 2) = 0$.

$\boxed{w_0 = -2}$ ⇒ fixed the bias.

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Calculate or che

$y_i(x_1w_1 + x_2w_2 + x_3w_3$

$\geq 1 \rightarrow$ Perfect classifica

Now,

$f(x) = w_1x_1 + w_2x_2 + w_3x_3 - 2.$

Check $y_i(x_i - 2)$.

$f(x) = y_i(x_i - 2) = 0.$ $\hat{y}$

① $1(6-2) = 4 \geq 1.$

② $-1(1-2) = 1 \geq 1$

③ $1(4-2) = 2 \geq 1$

④ $-(0-2) = 2 \geq 1$

⑤ $1(5-2) = 3 \geq 1.$

Cost or Loss fn

$$\boxed{L = \frac{1}{2}\|w\|^2 - \sum_i C_i\left[y_i(wx_i + b) - 1\right]}$$

$$= \frac{1}{2}(w_1^2 + w_2^2 + w_3^2) - \sum_{i=1}^{N} C_i \text{ Error}$$

Update the weights.

If P.F $\Rightarrow y_i(w_1x_1 + w_2x_2 + \dots) \geq 1$. then

$C \rightarrow 0. \rightarrow$ Error value should be low

$$\boxed{w^{new} = w^{old} - \eta \frac{\partial L}{\partial w}}$$

$\left(C \rightarrow 0 \rightarrow L = \frac{1}{2}w^2.\right)$

$$= \boxed{w^{old} - \eta w^{old}}$$

Quick Work

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$$\boxed{w_1^{new} = w_1^{old} - \eta\, w_1^{old}} \rightarrow \text{weights.}$$

$$\boxed{w_1^{new} = w_0^{old} - \eta\, w_0^{old}} \rightarrow \text{bias.}$$

$w_1 = 1 - (0.1)(1)$

$w_1^{new} = 1 - 0.1$

$$\boxed{w_1^{new} = 0.9}$$

$$\text{Margine} = \frac{2}{\|w\|} = \frac{2}{\sqrt{w_1^2 + w_2^2 + w_3^2 + \dots}}$$

(∵ Margine should be increase)

$\Rightarrow$ If N.P.F $\Rightarrow y_i(w_1x_1 + w_2x_2 + \dots) \leq 1$. | Take C value.

$$w^{new} = w - \eta \frac{\partial L}{\partial w}$$

$$= \boxed{w - \eta\left[w - \sum_i C_i y_i x_i\right]}$$

$$= \boxed{(1-\eta)w + \eta \sum_i C_i y_i x_i}$$

$w_1 = \sum_i C_i y_i x_i = C \sum_i y_i x_i$

$C = 1 \Rightarrow (y_1 x_{11} + y_2 x_{12} + y_3 x_{13} + y_4 x_{14} + y_5 x_{15})$